

Practical Lecture 1

McCulloch-Pitts Neurons and Logic Functions

MATLAB Laboratory



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Implement AND, OR, NAND, NOR, and NOT with fixed weights and a threshold activation.

The McCulloch-Pitts neuron calculates a weighted sum, $net = Xw$, then returns 1 when the net input reaches the selected threshold. The experiment shows how different weights and thresholds reproduce standard truth tables without a training stage.

Expected learning outcome: students connect the mathematical rule with the values printed by the MATLAB program.



- 1 Create every binary input combination.
- 2 Store the weights and threshold for each gate.
- 3 Calculate the weighted sum for each input row.
- 4 Apply the hard threshold and display the truth table.



```
%% Practical 1: McCulloch-Pitts neurons and logic functions
clear; clc;

X = [0 0; 0 1; 1 0; 1 1];

gates = {
    'AND', [ 1 1], 2;
    'OR', [ 1 1], 1;
    'NAND', [-1 -1], -1;
    'NOR', [-1 -1], 0
};

for g = 1:size(gates,1)
    name = gates{g,1};
    weights = gates{g,2};
    threshold = gates{g,3};
    output = mpNeuron(X, weights, threshold);
end
```



```
fprintf('\n%s gate\n', name);
disp(table(X(:,1), X(:,2), output, ...
    'VariableNames', {'x1','x2','y'}));
end

xNot = [0; 1];
yNot = mpNeuron(xNot, -1, 0);
fprintf('\nNOT gate\n');
disp(table(xNot, yNot, 'VariableNames', {'x','y'}));

function y = mpNeuron(X, weights, threshold)
% Calculate the weighted sum for every input row.
net = X * weights(:);
% Apply the binary threshold activation.
y = double(net >= threshold);
end
```



MATLAB Command Window

```
>> Practical_01_McCulloch_Pitts_MATLAB
```

```
AND gate:  0 0 0 1
```

```
OR gate:   0 1 1 1
```

```
NAND gate: 1 1 1 0
```

```
NOR gate:  1 0 0 0
```

```
NOT gate:  1 0
```